MASTERS CAPSTONE THESIS PROPOSAL

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Ethical Evaluation of AI Agents: A Systematic Audit of Ethical Gaps in AI Agent Evaluation and a Proposed Ethical Evaluation Framework

Research Question

What ethical dimensions are systematically absent from current AI agent benchmarks, and how can a structured ethical evaluation framework address those gaps?

Background & Motivation

AI agents (autonomous systems capable of browsing the web, executing code, sending communications, and chaining multi-step tasks with minimal human oversight) are rapidly transitioning from research settings into real-world deployment. Current evaluation frameworks such as WebArena, AgentBench, and SWE-bench assess agents primarily on task completion, efficiency, and accuracy. Existing ethics benchmarks (MoralBench, ETHICS) target language model outputs rather than agent actions, while safety-focused agent benchmarks (AgentSafetyBench, OS-HARM) address only a narrow slice of ethical behavior, leaving critical dimensions such as autonomy preservation, deception, privacy, and accountability entirely unmeasured. No work currently audits the full landscape of agent benchmarks through a comprehensive ethical lens, nor proposes a validated framework to address those gaps. This thesis addresses that gap directly.

Gap in Existing Literature

- Ethics benchmarks test what agents say; none systematically evaluate what they do
- Safety benchmarks for agents cover misuse and jailbreaks, not comprehensive ethical behavior
- No audit of existing agent benchmarks through a multi-dimensional ethical rubric exists in the literature
- No validated ethical evaluation framework designed specifically for AI agents has been proposed

Proposed Methodology

This thesis employs a three-phase mixed methods approach:

- Phase 1: Systematic Literature Review: Comprehensive mapping of existing agent benchmarks and AI ethics frameworks to identify the current state and boundaries of the field.
- Phase 2: Empirical Benchmark Audit: A structured ethical rubric, grounded in established AI ethics principles (autonomy, beneficence, non-maleficence, justice, transparency, accountability), is applied to 8–12 existing agent benchmarks. Inter-rater reliability testing ensures rubric validity.
- Phase 3: Framework Proposal & Validation: An Ethical Evaluation Framework for AI Agents is proposed and empirically validated by applying it to 2–3 real deployed agents (e.g., AutoGPT, Claude Computer Use) to demonstrate it surfaces ethical issues that existing benchmarks miss.

Expected Contributions

- First systematic audit of AI agent benchmarks revealing ethical blind spots beyond safety compliance
- A novel, operationalizable Ethical Evaluation Framework designed specifically for AI agents
- Empirical validation demonstrating the framework surfaces issues that existing benchmarks miss
- A reusable rubric and methodology for ongoing ethical evaluation of future agent benchmarks
- Policy-relevant recommendations for benchmark developers, AI regulators, and deployers